

Practice Midterm II with Solutions

GSE 510

Total Points: 100
10 Questions, 10 Points Each

Instructions: Answer all questions. Show your work clearly. Each question is worth 10 points. Partial credit will be given for correct reasoning.

Practice Midterm II

1. Sets, Logic, and Budget Sets

(10 points)

Let

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : 3x_1 + 2x_2 \leq 12\}.$$

Answer the following:

- (a) Is $(2, 2) \in B$?
- (b) Is $(4, 1) \in B$?
- (c) Find the largest possible value of x_1 in B .
- (d) Find the largest possible value of x_2 in B .
- (e) Is B closed and bounded? Briefly explain.

2. Sequences and Limits

(10 points)

For each sequence, determine whether it converges. If it converges, find the limit.

$$a_n = 4 - \frac{2}{n}, \quad b_n = \frac{3n+5}{2n}, \quad c_n = (-1)^n \frac{1}{n}, \quad d_n = (-1)^n.$$

3. Open, Closed, and Boundary Points

(10 points)

Consider the set

$$S = [0, 2).$$

- (a) Is S open in $\mathbb{R}$?
- (b) Is S closed in $\mathbb{R}$?
- (c) Find the boundary points of S .
- (d) Give a sequence in S that converges to a point not in S .
- (e) Give a sequence in S that converges to a point in S .

4. Linear Algebra and Systems

(10 points)

Consider the system

$$3x_1 + x_2 = 13, \quad 2x_1 + 4x_2 = 20.$$

- (a) Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.

- (b) Compute $|\mathbf{A}|$.
- (c) Is $\mathbf{A}$ nonsingular?
- (d) Solve for x_1 and x_2 .
- (e) What does nonsingularity imply about the number of solutions?

5. Linear Independence and Span

(10 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

- (a) Write a general linear combination $a\mathbf{v}_1 + b\mathbf{v}_2$.
- (b) Are $\mathbf{v}_1$ and $\mathbf{v}_2$ linearly independent?
- (c) Do $\mathbf{v}_1$ and $\mathbf{v}_2$ span $\mathbb{R}^2$?
- (d) Can they generate the vector

$$\begin{bmatrix} 6 \\ 9 \end{bmatrix}?$$

If yes, give one pair (a, b) that works.

- (e) Can they generate the vector

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}?$$

Explain.

6. Regression in Matrix Form

(10 points)

Suppose we estimate

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

using

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}.$$

- (a) Write the regression model in matrix form.
- (b) Compute $\mathbf{X}'\mathbf{X}$.
- (c) Compute $\mathbf{X}'\mathbf{y}$.
- (d) Compute $(\mathbf{X}'\mathbf{X})^{-1}$.
- (e) Compute $\hat{\boldsymbol{\beta}}$.
- (f) Write the fitted regression line.

7. Derivatives and Economic Interpretation

(10 points)

Suppose a firm's total cost function is

$$C(q) = 40 + 8q + 2q^2.$$

- (a) Find the marginal cost function.
- (b) Find $C'(10)$.
- (c) Find $C''(q)$.

- (d) Is marginal cost increasing or decreasing?
- (e) Use a linear approximation to estimate $C(11) - C(10)$.
- (f) Compute the exact value of $C(11) - C(10)$.

8. Optimization

(10 points)

A firm has revenue

$$R(q) = 90q - 3q^2$$

and cost

$$C(q) = 30 + 15q + q^2.$$

Profit is $\pi(q) = R(q) - C(q)$.

- (a) Write $\pi(q)$.
- (b) Find $\pi'(q)$.
- (c) Solve the first-order condition.
- (d) Find $\pi''(q)$.
- (e) Is the critical point a maximum or minimum?
- (f) Find maximum profit.

9. Approximation

(10 points)

Suppose output is produced according to

$$Y = F(K) = 15K^{1/2}.$$

The current capital stock is $K_0 = 64$.

- (a) Find $F'(K)$ and $F''(K)$.
- (b) Compute $F(64)$.
- (c) Use a linear approximation around $K_0 = 64$ to approximate $F(68)$.
- (d) Use a quadratic approximation around $K_0 = 64$ to approximate $F(68)$.
- (e) Which approximation should be closer to the exact value? Explain briefly.

10. Implicit Differentiation and Comparative Statics

(10 points)

Suppose market equilibrium is described by

$$D(p, a) = S(p),$$

where

$$D(p, a) = a - p^2$$

and

$$S(p) = 4p.$$

Here a is a demand-shift parameter.

- (a) Write the equilibrium condition as $F(p, a) = 0$.
- (b) Compute F_p and F_a .

- (c) Use the implicit function theorem formula

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}$$

to find $\frac{dp^*}{da}$.

- (d) Evaluate $\frac{dp^*}{da}$ at $p = 2$.

- (e) Use a linear approximation to estimate the change in equilibrium price if a increases by 3 when the current equilibrium price is $p = 2$.

- (f) Interpret the sign of $\frac{dp^*}{da}$ economically.

Solutions

1. Sets, Logic, and Budget Sets

The budget set is

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : 3x_1 + 2x_2 \leq 12\}.$$

(a) For $(2, 2)$,

$$3(2) + 2(2) = 6 + 4 = 10 \leq 12.$$

Therefore, $(2, 2) \in B$.

(b) For $(4, 1)$,

$$3(4) + 2(1) = 12 + 2 = 14 > 12.$$

Therefore, $(4, 1) \notin B$.

(c) To maximize x_1 , set $x_2 = 0$:

$$3x_1 \leq 12.$$

Thus,

$$x_1 \leq 4.$$

The largest possible value of x_1 is 4.

(d) To maximize x_2 , set $x_1 = 0$:

$$2x_2 \leq 12.$$

Thus,

$$x_2 \leq 6.$$

The largest possible value of x_2 is 6.

(e) The set is closed because it is defined by weak inequalities. It is bounded because

$$0 \leq x_1 \leq 4, \quad 0 \leq x_2 \leq 6.$$

2. Sequences and Limits

• For

$$a_n = 4 - \frac{2}{n},$$

since $\frac{2}{n} \rightarrow 0$,

$$a_n \rightarrow 4.$$

• For

$$b_n = \frac{3n+5}{2n} = \frac{3}{2} + \frac{5}{2n},$$

since $\frac{5}{2n} \rightarrow 0$,

$$b_n \rightarrow \frac{3}{2}.$$

• For

$$c_n = (-1)^n \frac{1}{n},$$

the sign alternates, but the magnitude goes to zero. Hence

$$c_n \rightarrow 0.$$

• For

$$d_n = (-1)^n,$$

the sequence alternates between 1 and -1 . Therefore, it does not converge.

3. Open, Closed, and Boundary Points

Let

$$S = [0, 2).$$

- (a) The set is not open in $\mathbb{R}$ because it contains 0, but no open interval around 0 is contained entirely in S .
- (b) The set is not closed because it does not contain the boundary point 2.
- (c) The boundary points are

$$0 \quad \text{and} \quad 2.$$

- (d) One sequence in S converging to a point not in S is

$$x_n = 2 - \frac{1}{n}.$$

Then $x_n \in S$ for all n , but

$$x_n \rightarrow 2,$$

and $2 \notin S$.

- (e) One sequence in S converging to a point in S is

$$y_n = \frac{1}{n}.$$

Then $y_n \in S$ and

$$y_n \rightarrow 0,$$

with $0 \in S$.

4. Linear Algebra and Systems

The system is

$$3x_1 + x_2 = 13, \quad 2x_1 + 4x_2 = 20.$$

- (a) In matrix form:

$$\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 13 \\ 20 \end{bmatrix}.$$

- (b) The determinant is

$$|\mathbf{A}| = 3(4) - 1(2) = 12 - 2 = 10.$$

- (c) Since $|\mathbf{A}| = 10 \neq 0$, $\mathbf{A}$ is nonsingular.

- (d) From the first equation,

$$x_2 = 13 - 3x_1.$$

Substitute into the second:

$$2x_1 + 4(13 - 3x_1) = 20.$$

Therefore,

$$2x_1 + 52 - 12x_1 = 20.$$

Hence,

$$-10x_1 = -32,$$

so

$$x_1 = \frac{16}{5}.$$

Then

$$x_2 = 13 - 3\left(\frac{16}{5}\right) = \frac{65}{5} - \frac{48}{5} = \frac{17}{5}.$$

Thus,

$$x_1 = \frac{16}{5}, \quad x_2 = \frac{17}{5}.$$

(e) Nonsingularity implies that the system has a unique solution.

5. Linear Independence and Span

We have

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

(a) A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 2a + 4b \\ 3a + 6b \end{bmatrix}.$$

(b) Since

$$\mathbf{v}_2 = 2\mathbf{v}_1,$$

the vectors are linearly dependent. Therefore, they are not linearly independent.

(c) Since the vectors are multiples of each other, they lie on the same line. Therefore, they do not span $\mathbb{R}^2$.

(d) Yes. Since

$$\begin{bmatrix} 6 \\ 9 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

one solution is

$$a = 3, \quad b = 0.$$

(e) No. Any linear combination has the form

$$\begin{bmatrix} 2a + 4b \\ 3a + 6b \end{bmatrix} = (a + 2b) \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Therefore, every generated vector lies on the line where the second component is $\frac{3}{2}$ times the first component. The vector

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

does not lie on this line.

6. Regression in Matrix Form

The regression model is

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

(a) In matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

(b) We have

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}.$$

Therefore,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix}.$$

(c)

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 23 \end{bmatrix}.$$

(d) The determinant of $\mathbf{X}'\mathbf{X}$ is

$$3(21) - 7(7) = 63 - 49 = 14.$$

Thus,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{14} \begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix}.$$

(e) Therefore,

$$\hat{\beta} = \frac{1}{14} \begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 8 \\ 23 \end{bmatrix}.$$

Multiplying,

$$\begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 8 \\ 23 \end{bmatrix} = \begin{bmatrix} 168 - 161 \\ -56 + 69 \end{bmatrix} = \begin{bmatrix} 7 \\ 13 \end{bmatrix}.$$

Hence,

$$\hat{\beta} = \begin{bmatrix} 7/14 \\ 13/14 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 13/14 \end{bmatrix}.$$

(f) The fitted regression line is

$$\hat{y} = \frac{1}{2} + \frac{13}{14}x.$$

7. Derivatives and Economic Interpretation

The cost function is

$$C(q) = 40 + 8q + 2q^2.$$

(a) Marginal cost is

$$C'(q) = 8 + 4q.$$

(b)

$$C'(10) = 8 + 4(10) = 48.$$

(c)

$$C''(q) = 4.$$

(d) Since $C''(q) = 4 > 0$, marginal cost is increasing.

(e) A linear approximation gives

$$C(11) - C(10) \approx C'(10)(11 - 10).$$

Thus,

$$C(11) - C(10) \approx 48.$$

(f) The exact difference is

$$C(11) - C(10).$$

First,

$$C(11) = 40 + 8(11) + 2(11)^2 = 40 + 88 + 242 = 370.$$

Also,

$$C(10) = 40 + 80 + 200 = 320.$$

Therefore,

$$C(11) - C(10) = 370 - 320 = 50.$$

8. Optimization

We have

$$R(q) = 90q - 3q^2$$

and

$$C(q) = 30 + 15q + q^2.$$

(a) Profit is

$$\pi(q) = R(q) - C(q) = 90q - 3q^2 - (30 + 15q + q^2).$$

Therefore,

$$\pi(q) = 75q - 4q^2 - 30.$$

(b) The derivative is

$$\pi'(q) = 75 - 8q.$$

(c) The first-order condition is

$$75 - 8q = 0.$$

Hence,

$$q^* = \frac{75}{8}.$$

(d) The second derivative is

$$\pi''(q) = -8.$$

(e) Since $\pi''(q) < 0$, the critical point is a local maximum.

(f) Maximum profit is

$$\pi\left(\frac{75}{8}\right) = 75\left(\frac{75}{8}\right) - 4\left(\frac{75}{8}\right)^2 - 30.$$

Compute:

$$75\left(\frac{75}{8}\right) = \frac{5625}{8},$$

and

$$4\left(\frac{75}{8}\right)^2 = 4\left(\frac{5625}{64}\right) = \frac{5625}{16}.$$

Therefore,

$$\pi\left(\frac{75}{8}\right) = \frac{5625}{8} - \frac{5625}{16} - 30 = \frac{5625}{16} - 30.$$

Since

$$30 = \frac{480}{16},$$

we get

$$\pi\left(\frac{75}{8}\right) = \frac{5145}{16}.$$

9. Approximation

The production function is

$$F(K) = 15K^{1/2}.$$

(a) The first derivative is

$$F'(K) = \frac{15}{2}K^{-1/2}.$$

The second derivative is

$$F''(K) = \frac{15}{2}\left(-\frac{1}{2}\right)K^{-3/2} = -\frac{15}{4}K^{-3/2}.$$

(b)

$$F(64) = 15(64)^{1/2} = 15(8) = 120.$$

(c) Since

$$F'(64) = \frac{15}{2}(64)^{-1/2} = \frac{15}{2} \cdot \frac{1}{8} = \frac{15}{16},$$

and

$$68 - 64 = 4,$$

the linear approximation gives

$$F(68) \approx F(64) + F'(64)(68 - 64).$$

Thus,

$$F(68) \approx 120 + \frac{15}{16}(4) = 120 + \frac{60}{16} = 120 + \frac{15}{4} = 123.75.$$

(d) First,

$$F''(64) = -\frac{15}{4}(64)^{-3/2}.$$

Since

$$64^{3/2} = 512,$$

we have

$$F''(64) = -\frac{15}{4} \cdot \frac{1}{512} = -\frac{15}{2048}.$$

The quadratic approximation is

$$F(68) \approx F(64) + F'(64)(4) + \frac{1}{2}F''(64)(4)^2.$$

Therefore,

$$F(68) \approx 120 + \frac{15}{16}(4) + \frac{1}{2}\left(-\frac{15}{2048}\right)(16).$$

Simplifying,

$$F(68) \approx 123.75 - \frac{120}{2048}.$$

Thus,

$$F(68) \approx 123.75 - 0.05859375 = 123.69140625.$$

(e) The quadratic approximation should be closer because it includes information about curvature through the second derivative.

10. Implicit Differentiation and Comparative Statics

Equilibrium is defined by

$$D(p, a) = S(p),$$

where

$$D(p, a) = a - p^2$$

and

$$S(p) = 4p.$$

(a) The equilibrium condition is

$$a - p^2 = 4p.$$

Bring all terms to one side:

$$F(p, a) = a - p^2 - 4p = 0.$$

(b) We compute

$$F_p = -2p - 4$$

and

$$F_a = 1.$$

(c) By the implicit function theorem,

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}.$$

Therefore,

$$\frac{dp^*}{da} = -\frac{1}{-2p-4} = \frac{1}{2p+4}.$$

(d) At $p = 2$,

$$\frac{dp^*}{da} = \frac{1}{2(2)+4} = \frac{1}{8}.$$

(e) If a increases by 3, then

$$\Delta a = 3.$$

Using linear approximation,

$$\Delta p^* \approx \frac{dp^*}{da} \Delta a = \frac{1}{8}(3) = \frac{3}{8}.$$

So the equilibrium price increases by approximately

$$\frac{3}{8} = 0.375.$$

(f) Since

$$\frac{dp^*}{da} > 0,$$

an increase in the demand-shift parameter raises equilibrium price. Economically, demand shifts outward, putting upward pressure on price.